



2.3 Exponential and logarithmic models

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

An exponential model $N(t) = N_0 b^t$ or $N(t) = N_0 e^{(kt)}$ describes multiplicative growth ($b > 1$ or $k > 0$) or decay ($0 < b < 1$ or $k < 0$) from an initial value N_0 , with a horizontal asymptote at $N = 0$ (or a shifted value). Logarithms are used to solve for the exponent, e.g. to find a doubling time or half-life. Such models suit contexts like population growth, radioactive decay, and depreciation, where change is proportional to the current amount.

Worked example: follow the reasoning

A radioactive sample decays according to $N(t) = 200e^{(-0.05t)}$ grams, t in years. Find (a) the mass after 10 years, (b) the half-life.

Solution: (a) $N(10) = 200e^{(-0.5)} = 200 \times 0.6065 \approx 121.3$ g. (b) $200e^{(-0.05t)} = 100 \rightarrow e^{(-0.05t)} = 0.5 \rightarrow t = \ln(0.5)/(-0.05) = 13.86$ years.

Quick check · multiple choice

Which function models exponential decay?

- A $y = 3(1.2)^x$
- B $y = 3(0.8)^x$
- C $y = 3x^2$
- D $y = 3 + 0.8x$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The number of bacteria in a culture is modelled by $N(t) = 500e^{(0.18t)}$, where t is in hours. (a) Find the number of bacteria after 6 hours, to the nearest whole number. (b) Find the time taken for the population to reach 5000, to the nearest 0.1 hour. (c) State the initial population and explain one limitation of this model for very large t . [7]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $N(6)=500e^{(1.08)}=500 \times 2.9447 \approx 1472$ bacteria. (b) $500e^{(0.18t)}=5000 \rightarrow e^{(0.18t)}=10 \rightarrow t=\ln 10/0.18=2.302585/0.18 \approx 12.8$ hours. (c) Initial population $N(0)=500$; as $t \rightarrow \infty$, $N(t) \rightarrow \infty$, which is unrealistic since real populations are eventually limited by resources, so the model only applies over a limited time range.

Quick check answer

B. A base strictly between 0 and 1 (0.8) gives decreasing exponential values as x increases.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.